

Arterial embolisation for kidney cancer

Arterial embolisation is when a substance is injected into a blood vessel (artery) in the kidney. This can shrink the tumour by stopping the blood supply.

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What is arterial embolisation?

Arterial embolisation is when a substance is injected into a blood vessel (artery) in the [kidney](#), to block the blood supply to the cancer. This reduces the supply of oxygen and nutrients to the tumour, which can make it shrink or stop growing.

Arterial embolisation cannot cure the cancer by itself. But it can help control [symptoms](#), such as pain or bleeding.

You may be offered arterial embolisation if an operation is not possible for you.

Having arterial embolisation for kidney cancer

Before treatment, the nurse or doctor usually gives you a mild sedative to help you relax. They then inject some local anaesthetic into the skin at the top of the leg (the groin) to numb the area.

A doctor puts a thin, plastic tube (catheter) into a blood vessel in the groin. They guide the catheter using the x-ray images to help. They continue upwards until the tip reaches the artery that carries blood to the area of the kidney with the cancer. They then inject a substance, such as tiny beads, through the catheter into the artery. This blocks the blood supply to the cancer.

Image: The illustration shows how tumour embolisation is done for a kidney cancer. It shows the chest and abdomen (trunk) of a person, containing 2 kidneys. At the centre of the right kidney is a tumour. Also shown are the major blood vessels that connect the kidneys and flow down the legs through the groin. A thin plastic tube (catheter) has been put into the blood vessel in the right side of the groin. The catheter leads up through the blood vessel and into the right kidney. Attached to the catheter, outside of the body, is a syringe that contains the substance used to block the blood supply to the tumour.

Image: The illustration shows a close-up of tumour embolisation for kidney cancer. It shows a kidney with a tumour in its centre. To the side of the kidney is the large blood vessel that carries blood to the kidney. Inside the blood vessel is a thin catheter. The tip of the catheter sits in the blood vessel just outside the kidney. Inside the catheter are tiny embolisation beads that have been injected through the catheter to block the small blood vessels that carry blood to the tumour.

Side effects of arterial embolisation for kidney cancer

This treatment can sometimes cause pain in the back. Your doctor will give you painkillers to take regularly for a few days.

You may feel a little unwell for the first few days and have a slightly raised temperature. You will probably also feel tired.

Going home after arterial embolisation

After arterial embolisation, you may need to stay in hospital overnight, or for 1 or 2 days. This depends on how quickly you recover.

Your surgeon or nurse can tell you when you can start doing everyday activities again.

Before you leave hospital, you will be given an appointment for a check-up. This will be at an outpatient clinic. Your doctor or nurse may also arrange for you to have a [CT scan](#) to see the result of the treatment.

You will have regular follow-up scans to check the kidney for any signs of the cancer growing.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our kidney cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

Ljunberg B, Albiges L, Bedke J et al. European Association of Urology Guidelines on renal cell carcinoma. 2023. Available from <https://uroweb.org/guidelines/renal-cell-carcinoma>

Powles T, Albiges L, Bex A, et al. Renal cell carcinoma: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Ann Oncol. 2024 May 22. Available from: <https://doi.org/10.1016/j.annonc.2024.05.537>

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help